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Piyush Arora

Professional Summary

Graduate student in MSc Computing (AI & ML) at Imperial College London, with a B.Tech in Computer Science and Engineering from IIT Jodhpur. Experienced as a **Research Engineer** and **Researcher** in building and evaluating VLMs, Speech and Audio-Visual models, including **LLM-based** audio-visual speech recognition and noise-robust representation learning using Variational Information Bottleneck. Currently exploring **Neural Codec Architectures**, including Matryoshka based codec design, for scalable and efficient speech representation. Published at **WACV** and **EMNLP**, with hands-on experience in **RL-based post-training** (GRPO), reproducible PyTorch pipelines on GPU clusters, and end-to-end model development across research and production settings.

Education

- **Imperial College London** — United Kingdom
  MSc Computing (Artificial Intelligence and Machine Learning); Term 1 Grade: Distinction — *Sep 2025 - Present*
- **Indian Institute of Technology Jodhpur** — India
  Bachelor of Technology in Computer Science and Engineering; CGPA: 8.88/10 — *Dec 2020 - May 2024*
  - *All India Rank **3316 (IIT-JEE 2020)** among 1.3 million applicants.*
  - *Selected for the prestigious **Mitacs Globalink Research Internship** and completed a funded summer research internship.*
  - *Capstone Project: Hybrid Sample Synthesis-Based Debiasing in Limited Data Settings published at **WACV 2024**.*
  - *Ranked among the top 5 students in the cohort.*
  - *Coursework: Deep Learning, Advanced Machine Learning, Natural Language Understanding, Data Structures and Algorithms, Operating Systems, Computer Networks, Probability, Statistics, Stochastic Processes, Linear Algebra.*

Publications

- **P. Arora***, N. Singh*, U. Cappellazzo, S. Petridis, and M. Pantic, *VIB-AVSR: Variational Information Bottleneck for Noise-Robust LLM-Based Audio-Visual Speech Recognition*: **INTERSPEECH 2026**
- **P. Arora***, N. Singh*, V. Diwan*, and P. Mazumder, *BLADE: Bias-Linked Adaptive DEbiasing*: Under Review
- A. Subramanyam*, N. Singh*, **P. Arora***, and A. Mishra, *When Big Models Train Small Ones: Label-Free Model Parity Alignment for Efficient Visual Question Answering with Small VLMs*: **EMNLP 2025**
- **P. Arora*** and P. Mazumder*, *Hybrid Sample Synthesis-Based Debiasing Of Classifier In Limited Data Setting*: **WACV 2024**

* denotes equal contribution

Work Experience

- **Neuphonic** — London, UK
  Machine Learning Intern — *May 2026 - Aug 2026*
  - Distilling LLM based noise-robust Automatic Speech Recognition (ASR) and audio language models for real-time offline deployment in noisy environments.
  - Building and optimising low-latency inference pipelines for efficient deployment on edge and GPU-based systems.
- **MetaFusion** — Noida, India
  Research Engineer — *May 2024 - Aug 2025*
  - Built end-to-end fine-tuning and evaluation pipelines for VLM/LLM-based surveillance captioning and attribute prediction in PyTorch; designed attribute-specific cross-attention layers, profiled/optimised bottlenecks, and fine-tuned Microsoft Florence-2, improving F1 score from **84% to 91%**.
  - Shipped an intelligent traffic management system integrating multi-violation detection, vehicle recognition, license plate OCR, and CLIP-based attribute prediction, deployed as containerised microservices on GPU-backed infrastructure.
  - Owned production rollout across Gandhinagar, Vizag, and Maharashtra with Gradio interfaces for real-time monitoring; built automated benchmarking for quality/latency/throughput and achieved **50%+** end-to-end throughput improvement.

Research Experience

- **Improving Chain-of-Thought Faithfulness in LLMs** — Imperial College London
  Supervisor: Dr. Oana-Maria Camburu | Code — *Oct 2025 – Feb 2026*
  - Developed **CIRL-Faith**, a concept-intervention based training algorithm that exposes and corrects a core AI safety failure: models producing reasoning chains misaligned with their internal computations, a risk that compounds in high-stakes deployments.
  - Formulated RL-based post-training using **Group Relative Policy Optimization (GRPO)** with intervention and process driven reward signals, reducing reliance on human filtering and enabling scalable oversight of model reasoning without multi-model inference.

- Evaluated across **Qwen3-4B, Qwen3-8B, and Llama3.1-8B** using established faithfulness metrics, achieving gains of up to **30%** over respective base models, with systematic ablations across model scales.

- **VIB-AVSR for Noise-Robust LLM-Based AVSR (Under Review)** — Imperial College London
  Supervisor: Dr. Umberto Cappellazzo, Dr. Stavros Petridis, Dr. Maja Pantic — *Nov 2025 – Mar 2026*
  - Developed **VIB-AVSR**, an LLM-based audio-visual speech recognition model connecting pre-trained audio-visual encoders to an LLM backbone, achieving strong performance in clean acoustic conditions.
  - Integrated Variational Information Bottleneck layers at targeted positions within the LLM backbone to regularize representations, addressing the lack of explicit noise-robustness mechanisms in existing LLM-based AVSR models.
  - Reduced performance degradation across multiple SNR levels and noise types without architectural modifications or additional training data, while maintaining efficient GPU inference across varying noise conditions.

- **Matryoshka-Based Neural Codec for Adaptive Speech Tokenization** — Imperial College London
  Supervisor: Dr. Mirco Ravanelli, Dr. Umberto Cappellazzo, Dr. Stavros Petridis — *Apr 2026 – Present*
  - Developing a Matryoshka-based neural codec for scalable speech tokenization, based on Binary Spherical Quantization, achieving adaptive token rates ranging from 6.25 to 50 tokens per second to enable flexible rate-quality tradeoffs at inference time.
  - Achieves comparable performance with strong baselines on standard speech benchmarks including LibriSpeech and LibriTTS, without sacrificing efficiency at lower token rates.

- **StyleLip: Cross-Identity Speaking Style Transfer (Under Review)** — Imperial College London
  Supervisors: Dr. Antoni Bigata, Dr. Stavros Petridis, Dr. Maja Pantic — *Oct 2025 – Mar 2026*
  - Developed StyleLip, a few-shot framework for speaking style transfer in audio-driven lip-synchronization, capturing and reproducing person-specific idiosyncratic lip movements across different identities using LoRA layers integrated into a base lip-sync model.
  - Regularized training via synthetically generated multi-identity samples with a shared speaking style to prevent identity leakage while enforcing consistent style transfer, achieving SOTA performance as a plug-and-play solution.
  - Introduced SSCS, a novel quantitative metric to measure speaking style transfer fidelity, addressing the lack of dedicated metrics in the field.

- **When Big Models Train Small Ones** — IIT Jodhpur
  Supervisor: Dr. Anand Mishra (VL2G Lab) | Technical Report — *Dec 2024 – Mar 2025*
  - Designed **Model Parity Alignment (MPA)** for Visual Question Answering to train small VLMs (SmolVLM-500M, TinyLLaVA-2B, InternVL2-2B/4B) from large VLMs (Qwen2VL-7B, InternVL2-8B, GPT-4o) using unlabeled data, implementing efficient PyTorch training pipelines across multiple model scales.
  - Developed disparity-aware training to mitigate hallucinations and close reasoning gaps between VLM and its larger counterpart via automated QA pair generation and filtering of disparity samples to guide fine-tuning across OCR, commonsense, and factual tasks.
  - Achieved consistent accuracy gains up to **6%** on VQA benchmarks like TextVQA, ST-VQA, OKVQA, MedicalVQA, and ChartQA.

- **Hybrid Sample Synthesis-Based Debiasing in Limited Data Settings** — IIT Jodhpur
  Supervisor: Dr. Pratik Mazumder | Technical Report — *Dec 2022 – Mar 2023*
  - Designed a hybrid sample synthesis method for low-data regimes, enabling ResNet classifiers to generalize under unknown biases and promoting AI fairness.
  - Built a dual-model framework to create bias-conflicting samples without relying on explicit bias annotations.
  - Achieved up to **+10%** accuracy improvement over prior methods on benchmarks such as corrupted CIFAR-10, Colored MNIST, and BFFHQ.

Skills Summary

- **Programming**: Python, C++, SQL, Bash
- **ML & AI Frameworks**: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV
- **LLM Post-Training & Evaluation**: Supervised Fine-Tuning (SFT), RL-based training (GRPO), Reward-Driven Optimization, Ablations, Benchmarking
- **NLP & Multimodal**: Transformers, Instruction Tuning, Reasoning Faithfulness; Audio-Visual Modeling
- **Quantitative Skills**: Probability Theory, Statistics, Optimization, Stochastic Processes, Monte Carlo Simulation
- **Tools & Platforms**: Docker, Kubernetes, Git, Linux; experience deploying ML services on GPU-based infrastructure
- **Other**: Technical Writing, Public Speaking, Leadership, Project Management